Mathematical Physics I (Fall 2026): Midterm Examination

Oct. 24, 2026

[total 20 pts, closed book/cellphone, no calculator, 90 minutes]

• First, make sure you have all 6 answer sheets. Write down your name and student ID on each of all 6 answer sheets. Then, number the sheets from (1) to (6) on the top right corner. Your answer to each problem must *only* be in the sheet with the matching number (e.g., your answer to Problem 2 must *only* be in sheet (2)). After the exam, you will separately turn in all 6 answer sheets, even if some sheets are still blank.

• Make sure you have all 6 problems. Have a quick look through them all and portion your time wisely. If you find any issue or question, you *must* raise it in the first 30 minutes. You have to stay in the room for that 30 minutes even if you have nothing to write down.

• Make your writing easy to read. Illegible answers will *not* be graded.

1. (a) [1 pt] Find the first three terms of the Maclaurin series for the following function:

$$\int_0^x \frac{dt}{\sqrt{1+t^2}}.$$

(b) [2 pt] Consider a state of a system (e.g., a particle in a box, or a quantized oscillator) whose energy when occupied by a single particle is ϵ_0. In the quantum statistical distribution of bosons with unspecified total number of particles (so-called *Bose-Einstein* statistics), the expected total energy of all the particles in this state is found to be

$$\bar{\epsilon} = \frac{\sum_{n=1}^{\infty} n\epsilon_0 e^{-\frac{n\epsilon_0}{k_B T}}}{\sum_{n=0}^{\infty} e^{-\frac{n\epsilon_0}{k_B T}}}, \tag{1}$$

where k_B is the Boltzmann constant, and T is the temperature. Show that the ratio becomes

$$\bar{\epsilon} = \frac{\epsilon_0}{e^{\frac{\epsilon_0}{k_B T}} - 1}$$

by first identifying Eq.(1)'s denominator as a binomial expansion of $\frac{1}{1-x}$ with $x = e^{-\frac{\epsilon_0}{k_B T}}$, and its numerator as a constant times $\frac{x}{(1-x)^2}$. Using a power series expansion, show also that $\bar{\epsilon}$ reduces to $k_B T$ in the classical limit of $k_B T \gg \epsilon_0$.

2. (a) [2 pt] Show that $\arcsin z = -i \ln (iz \pm \sqrt{1 - z^2})$, where z is a complex number.

(b) [1 pt] Using the *principal value* of the complex logarithm, evaluate the expression $i^{\ln i}$ in simple $x + iy$ form.

3. (a) [2 pt] Solve the set of equations below by <u>*all three*</u> methods listed here: *(i)* by row reducing the augmented matrix, *(ii)* by using Cramer's rule, *(iii)* by finding the inverse of the coefficient matrix. Clearly indicate which method you are using for each part of your answer, so that the grader could follow it easily.

$$\begin{cases} 2x - y + z = 5 \\ x + 2y - z = 4 \\ 3x + y + 2z = 17 \end{cases}$$

(b) [2 pt] Verify that the following matrix H is Hermitian. Find its eigenvalues and eigenvectors, and write a unitary matrix U which diagonalizes H by a similarity transformation. Finally, show that $U^{-1}HU$ yields the diagonal matrix of eigenvalues.

$$H = \begin{pmatrix} -2 & 3 + 4i \\ 3 - 4i & -2 \end{pmatrix}$$

(c) [1 pt] Consider a particle moving in the xy-plane near the origin. For small displacements from the origin, suppose that the force acting on the particle is given by

$$\mathbf{F} = -\begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -M\mathbf{x}.$$

Notice that, in general, the x- and y-motion are *coupled* — e.g., a displacement along the x-axis does not produce a force purely in the x-direction. Now, consider a sufficiently small displacement $\mathbf{x} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$. Show that the force is parallel to the displacement only along certain special directions (so-called *principal* directions). Explain why this is an eigenvalue problem, and find all eigenvalues and their corresponding normalized eigenvectors. Finally, identify which principal direction experiences the stronger restoring force.

(d) [1 pt] Now consider a modified force law,

$$\mathbf{F} = -\begin{pmatrix} 4 & 5 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -M'\mathbf{x}.$$

Find the eigenvalues and their normalized eigenvectors for this new matrix. Show that one of the eigenvalues is negative and the other is positive, and explain what this means physically for the stability of the equilibrium at the origin.[1]

[1] This behavior is characteristic of a *saddle point* discussed in Boas Chapter 4, Section 8.

4. (a) [1 pt] Find the point on $2x + 3y + z - 12 = 0$ for which $2x^2 + y^2 + z^2$ is a minimum.

(b) [2 pt] If $x^2 + y^2 = 2st - 10$ and $2xy = s^2 - t^2$, find $\frac{\partial x}{\partial s}$, $\frac{\partial x}{\partial t}$, $\frac{\partial y}{\partial s}$, $\frac{\partial y}{\partial t}$ at $(x, y, s, t) = (4, 2, 5, 3)$.

5. [2 pt] Compute the gravitational force on a unit mass located at the origin, due to the mass of uniform density ρ occupying the volume inside the sphere $r = 2a$ (centered at the origin) and above the plane $z = a$. (Note: Make sure to specify both the magnitude and the direction of the resulting force. You may invoke symmetry arguments to explain why certain components of the net force are zero. The gravitational constant is G.)

6. (a) [1 pt] Prove that $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})$. Here you are asked to prove the identity by explicitly working with the components, i.e.,

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = \begin{vmatrix} (\mathbf{A} \times \mathbf{B})_x & (\mathbf{A} \times \mathbf{B})_y & (\mathbf{A} \times \mathbf{B})_z \\ C_x & C_y & C_z \\ D_x & D_y & D_z \end{vmatrix} = ...$$

You are welcomed to also tackle this problem using tensor notation for an additional +1 point.

(b) [1 pt] Evaluate the integral

$$\iint_\sigma (\nabla \times \mathbf{V}) \cdot \mathbf{n} d\sigma$$

over the entire surface σ of a pyramid whose base is the square in the (x, y)-plane with corners at $(0, 0)$, $(0, 2)$, $(2, 0)$, $(2, 2)$ and whose top vertex is at $(1, 1, 2)$, where $\mathbf{V} = (x^2 z - 2)\mathbf{i} + (x + y - z)\mathbf{j} - xyz\mathbf{k}$. Here, $\mathbf{n}$ denotes the unit normal vector to σ, pointing outward from the pyramid.

(c) [1 pt] Evaluate the integral

$$\iint_\sigma \mathbf{V} \cdot \mathbf{n} d\sigma$$

over the entire surface σ of the cylinder defined by $x^2 + y^2 = 16$, $z = 5$ and $z = -5$, where $\mathbf{V} = (y^2 - x^2)\mathbf{i} + (2xy - y)\mathbf{j} + 3z\mathbf{k}$. Here, $\mathbf{n}$ denotes the unit normal vector to σ, pointing outward from the cylinder.